Energy Reports 12 (2024) 418–429

Contents lists available at ScienceDirect

Energy Reports

journal homepage: www.elsevier.com/locate/egyr

Research paper

Residential PV-battery scheduling with stochastic optimization and neural network-driven scenario generation

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ARTICLE INFO

Keywords:
Battery energy storage system
Behind-the-meter systems
Photovoltaic
Stochastic optimization
Uncertain systems

ABSTRACT

Residential end-users have increasingly been attracted to installing behind-the-meter (BTM) photovoltaic (PV) and battery energy storage systems (BESS). The inherent stochastic nature of PV introduces challenges to energy management for PV-BESS owners. This work presents a two-stage stochastic optimization (SO) framework for battery operation that takes into account uncertainties in PV outputs to minimize daily grid consumption and battery degradation. A combination of a feed-forward neural network (FFNN) and an error analysis statistical technique is proposed to generate a highly accurate scenario set for estimating PV behavior. Then, the scenario set is reduced using a backward scenario reduction technique to address the issue of dimensionality that SO suffers from. The reduced scenario set is then fed into the two-stage SO model to calculate expected electricity costs and battery degradation for the day-ahead PV-BESS problem. The suggested model's performance in controlling BTM batteries is evaluated using real-world data collected from smart meters installed in a house in Aran Islands, Ireland. The results show that the model can effectively withstand sudden changes in the PV output and generate results that are comparable to an ideal case with accurate PV data available.

1. Introduction

Over the last few years, the installed capacity of roof-top photovoltaic (PV) systems has increased significantly. This trend is expected to continue, with distributed PV systems accounting for a significant portion of electricity generation by 2050 (Renewable Capacity Statistics, 2022). This can remarkably reduce costs and pollution levels by reducing the reliance on fossil fuels (Rezaeimozafar et al., 2022).

Huge penetration of stochastic PV, however, poses challenges to the demand-supply balance for microgrids. Hence, battery energy storage systems (BESS) have been introduced at various scales, ranging from small-scale for the power distribution system to medium/large-scale for the power transmission and generation systems (Yuan et al., 2022). Buildings with PV can install a BESS behind-the-meter (BTM) and benefit from demand load shifting, self-consumption, arbitrage activities, and peak demand reduction, depending on their host network regulations (Zhu et al., 2023). However, apart from regulations, the battery's efficacy can be influenced by the on-site PV variable behavior. For example, the authors in (Mahmud et al., 2020) show that unmanaged uncertainties in the PV-BESS problem can lead to increased battery charge/discharge cycles, ultimately resulting in a reduced battery lifespan. These two affecting factors have encouraged researchers to design models for the day-ahead BTM PV-BESS control problem, which can be divided into two major groups based on how PV is modeled.

The first group consists of deterministic approaches, including rule-based methods, as well as statistical and computer science techniques driven by historical data, which are used for predicting PV outputs. The authors in (Elloumi et al., 2023) propose a method to control the charge and discharge of batteries based on real-time PV output collected from smart meters. This method aims to maintain a balance between demand and supply at a minimum cost for households. In (Van Der Meer et al., 2018), an autoregressive integrated moving average model is used for PV estimation to minimize grid consumption for a commercial user. In (Khoury et al., 2015), a physical PV model is employed to address the PV-BESS problem, considering the long-term cost minimization. To control a home BESS, the authors in (Sorour et al., 2021) employ a hybrid PV predictive model and a real-time rule-based battery control model. The PV-BESS optimization model in (Dutta et al., 2017) is based

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https://doi.org/10.1016/j.egyr.2024.06.017
Received 30 January 2024; Received in revised form 27 May 2024; Accepted 8 June 2024
Available online 20 June 2024

on the persistence technique and considers PV outputs the same at each time step the following day as it is today. Although such techniques are fast and easy to implement, they are often unable to capture the dynamics of complex systems with non-linear behavior and sudden changes, both present in PV (Liu et al., 2018). Machine learning algorithms are seen as potential solutions for managing PV-BESS when dealing with unpredictable PV output (Elsisi et al., 2023; Farhoumandi et al., 2021).

The authors in (Bamisile et al., 2023) use two deep hybrid neural networks to predict day-ahead PV output. In (Mirhoseini and Ghaffarzadeh, 2020), a K-means clustering algorithm is used to estimate the PV profile for the PV-BESS problem in a grid-connected electric vehicle charging station. In (Pattanaik et al., 2024), the authors compare three different neural networks (MLP, RNN and LSTM) for day-ahead PV point forecasts, aiming to optimize BESS operation to minimize costs in a PV-BESS-thermal unit. An LSTM network is trained in (Qing and Niu, 2018) that uses online weather forecasting data to estimate a day-ahead PV for the BTM BESS problem. In (Vu and Chung, 2022), the authors propose a recurrent neural network-based predictive model to estimate the PV output to optimize energy dispatch strategies for stand-alone microgrids. Despite the remarkable results in PV prediction accuracy (Van Der Meer et al., 2018; Khoury et al., 2015; Sorour et al., 2021; Dutta et al., 2017; Liu et al., 2018; Elsisi et al., 2023; Farhoumandi et al., 2021; Bamisile et al., 2023; Mirhoseini and Ghaffarzadeh, 2020; Pattanaik et al., 2024), due to the omission of PV uncertainties, solutions derived from deterministic approaches may be suboptimal for the battery control problem (Shirsat and Tang, 2021). To this end, (Pascual et al., 2021; Arcos-Aviles et al., 2021) introduce multi-stage models for microgrids with thermal and electrical loads that can reduce the effects of inaccurate forecasts of renewables on real-time power exchange with the grid. However, real-time implementation of such models requires a committed computer to perform offline optimization for each stage, which increases the costs for low-power installations like households.

The second group consists of probabilistic methods and are defined as statistical analysis tools that estimate the probability of an event occurring again based on historical data. Unlike deterministic models, probabilistic models include elements of randomness which means that a probabilistic model is likely to produce different results even with the same initial conditions (Calafiore et al., 2011). Probabilistic optimization includes robust optimization (RO), distributionally RO, chance constraint optimization, and stochastic optimization (SO) (Riaz et al., 2022). The PV-BESS problem is commonly solved using RO and SO in the literature. RO is based on achieving the worst-case scenario, ensuring that the solution can withstand even extreme conditions. The PV-BESS problem for a residential unit is formulated in (Zhang et al., 2020) using a RO model. In (Ren et al., 2016), the authors define the size and range for the uncertain PV output to solve the BTM BESS scheduling problem using a RO framework. However, RO models may lead to conservative solutions which are inefficient for non-sensitive units, e.g., residential end-users (Bertsimas et al., 2011). Moreover, it necessitates detailed information about the uncertainty, such as its size and range, which is not easy to obtain for PV (Kazemzadeh et al., 2019). Stochastic optimization, on the other hand, calculates an expected value for an uncertain objective function based on a set of probabilistic scenarios (Chen et al., 2012). The scenario set represents the discrete realization of a random variable, and its size is often reduced before feeding into the objective function, to reduce computational complexity.

In SO, the quality of solutions is highly dependent on the scenario set. Scenario generation in many works is based on forecasts from linear regression models, which are incapable of estimating complex non-linear PV behavior. In other words, linear models are unable to capture the non-linear relationship between the output of a PV system and the factors that affect it, such as solar irradiance, temperature, and shading effects (Li et al., 2020). For instance, in (Maluenda et al., 2023), PV scenario generation relies on point forecasts from a linear regression model. Random values, derived from the mean and standard deviation of prediction errors, are added to these forecasts. However, since the original forecasts may lack accuracy, using values for the mean and standard deviation further reduces the accuracy of the generated scenarios. Hence, the authors in (Kaffash et al., 2021) propose a two-stage SO model based on PV scenarios calculated from point forecasts by a feed-forward neural network (FFNN). However, the network's design is not discussed and the prediction is only based on historical PV output, overlooking the impacts of environmental and date-time factors on PV behavior. Also, the accuracy of such a model is highly dependent on the amount of data available on the historical PV output. In the case of units with a short time history of PV panel usage, accurate PV estimates cannot be achieved due to the limited historical PV data available. In (Vagropoulos et al., 2016) a FFNN-based scenario generation model for stochastic variables in microgrids is presented, which is designed for estimations over very short time periods. However, they use the hyperbolic sigmoid function as the network's activation function, which is prone to the vanishing gradient problem for complex problems. Moreover, the model only forecasts one step ahead, which means for a day-ahead problem it must run repeatedly. Each step's forecast is based on the forecast from the preceding step, which leads to an exponential increase in the final step's prediction error. In (Yao et al., 2016), the authors use the sample average approximation technique by using Monte Carlo simulation to generate PV scenarios and a random sampling method is deployed to reduce the number of scenarios. But, the Monte Carlo method is unable to estimate highly stochastic PV, affecting the accuracy of solutions. PV scenarios are generated in (Huang et al., 2021) by feeding a variational autoencoder neural network with noisy data. The noise signal is generated by producing Gaussian distributed numbers constrained by user-supervised min/max values. However, their results indicate that the scenarios do not account for sudden, significant fluctuations in PV. Despite the aforementioned drawbacks, (Calafiore et al., 2011)- (Huang et al., 2021) demonstrate significant progress in addressing uncertainties in the PV-BESS problem. However, none incorporates battery degradation control in their models, which can potentially impact the effectiveness of their solutions.

Given the concerns mentioned above, in this work, a two-stage SO model employing an FFNN-based scenario generation technique is proposed for the PV-BESS problem. This model captures PV uncertainties in the day-ahead PV-BESS scheduling problem, by accurately modelling PV behavior through probabilistic scenarios. In this regard, PV scenarios are generated using a hybrid approach that combines an FFNN with a statistical error analysis model. This method considers the prediction error range across various hours, ensuring sudden changes in PV output are accounted for within the scenario set. Then, to reduce the computational complexity, a backward scenario reduction technique is used to reduce the number of scenarios while maintaining the accuracy of PV modelling. The problem is formulated as a mixed-integer linear programming (MILP) model to minimize expected daily electricity bill costs as well as battery degradation.

The main contributions of this work are as follows.

1) A combination of a two-stage scenario-based SO model with FFNN-driven PV scenario generation delivers accurate solutions comparable to a deterministic case with a perfect dataset. The objective is to minimize bill costs and the number of battery cycles, which is crucial for reducing battery degradation;
2) An FFNN is trained for day-ahead PV prediction, using historical PV output and meteorological data, reducing reliance solely on historical PV data and improving prediction accuracy. This method is especially useful in cases with limited historical PV data. Additionally, a time-wise error analysis of FFNN outputs ensures effective modelling of the PV system's dynamics and sudden changes in PV output throughout the day;
3) To emphasize the impact of uncertainty modeling on solutions, a thorough evaluation of the presented model is provided, comparing it to an ideal uncertainty-free case study, a rule-based model, and

both RO-based and scenario-based SO techniques proposed in the literature, using real-world data.

The remainder of this paper is organized as follows. Section 2 explains the scenario generation and scenario reduction techniques in detail. Section 3 corresponds to the formulation of the proposed two-stage SO model. Section 4 provides the simulation assumptions and optimization results. Section 5 concludes the paper.

2. PV generation uncertainty modeling

As discussed, uncertainties in a random variable are represented using a set of probabilistic scenarios in SO. In the PV-battery problem, each scenario represents one possible trajectory of the states of the PV system. The size of the scenario set is typically large since a PV has stochastic behavior that results in numerous scenarios. Therefore, it is recommended that the number of scenarios be decreased to simplify computation. The subsections below give a detailed overview of scenario generation and scenario reduction techniques proposed in this work.

2.1. Scenario generation

2.1.1. Day-ahead PV point forecasts

The first step in generating PV output scenarios is to obtain day-ahead point forecasts of PV power generation. To this end, many parametric and non-parametric probabilistic/statistical approaches are introduced in the literature. Apart from the benefits, the main challenge in using a parametric approach is the need to provide the probability distribution function (pdf) of the PV output profiles (Kazemzadeh et al., 2019). This requirement is unavailable or difficult to obtain because PV output does not follow any specific pdf due to its non-linear behavior, environmentally variability, and complex interactions between multiple factors (Kaffash et al., 2021). Therefore, it is difficult to characterize the PV output using a single mathematical model or distribution, necessitating the use of data-driven techniques like time series analysis or Machine Learning for modelling and prediction.

In this work, an artificial neural network (ANN) is used to forecast day-ahead PV power generation. ANNs are non-parametric methods for approximating non-linear relationships between user-provided input and output data sets (training data) to predict unseen data for complicated systems (Benghanem et al., 2009). This process, known as training an ANN, requires careful adjustment of the network's structure and hyperparameters including the number of hidden layers and neurons in each layer, type of activation function, training function, learning rate, batch size and epochs. These adjustments must be made while considering the specific application of the ANN. An ANN can have several layers depending on its application; however, in its most basic form, an ANN has three layers: one input layer, one hidden layer, and one output layer. More information about ANNs can be found in (An Introduction to Neural Networks, 2022).

In this study, a feed-forward neural network with the back-propagation algorithm is utilized for the PV forecasting problem (Huang et al., 2016). The architecture of the FFNN designed for this study is depicted in Fig. 1. The input layer consists of 41 neurons, $M = 41$, consisting of five environmental features and thirty-six DateTime features (12 months + 24 hours). The former set consists of beam irradiance on the inclined plane, diffuse irradiance on the inclined plane, reflected irradiance on the inclined plane, sun height, and air temperature. All of this information is publicly available for any given location in (JRC Photovoltaic Geographical Information System, 2022). The DateTime feature set includes the numbers for hours and the 12 months of the year. It is worth noting that the feature set is normalized between [0,1] before being fed into the input layer. The DateTime elements are normalized using the one-hot encoding method that converts categorical variables into binary vectors (Machine Learning with R, 2022). The main advantage of this method is its ability to preserve the distinctness of categories without imposing any ordinal relationship between them.

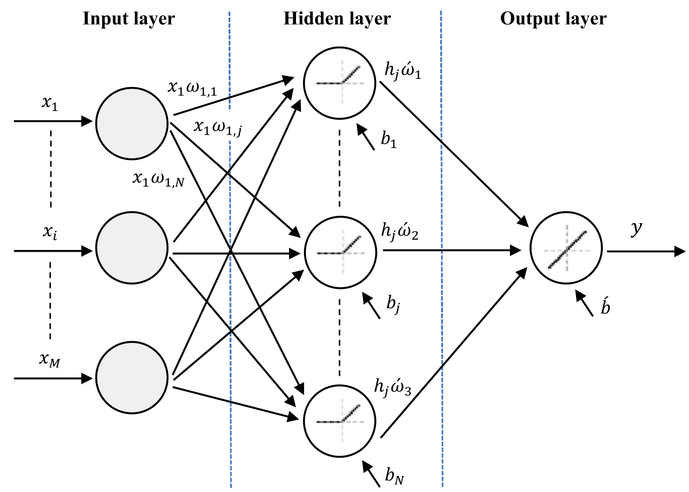

Fig. 1. The overall structure of the proposed FFNN for PV forecasting.

By incorporating environmental features in the network training process, the reliance solely on the PV output profile versus time is diminished. This allows the network to achieve reliable performance even when historical PV output data is relatively limited in size. Nevertheless, this raises the likelihood of the network encountering noisy data, as environmental features originate from monitoring sensors that may capture inaccurate values, leading to outliers in the dataset. However, this issue is addressed through data preprocessing, which involves removing outliers from the dataset. To this end, the Z-score, representing the number of standard deviations a data point deviates from the mean of the dataset, is calculated. A Z-score of 0 indicates alignment with the mean, with positive values above and negative values below. Outliers in this study are defined as data points with Z-scores exceeding +3 or falling below −3, and subsequently removed from the dataset.

The connection between FFNN layers is explained by (1) and (2) as follows (Huang et al., 2016).

$$h_j = f_H\left(\sum\nolimits_{i=1}^{M} x_i\omega_{i,j} + b_j\right), \quad j = 1:N \tag{1}$$

$$y = f_y\left(\sum\nolimits_{j=1}^{N} h_j\acute{\omega}_j + \acute{b}\right) \tag{2}$$

where x_i is the i^{th} input element, h_j is the j^{th} output of the hidden layer, and y is the output of the FFNN. Weights and biases are given by ω and b, respectively. f_H denotes the rectified linear unit (ReLU) activation function and f_y represents the linear activation function as transfer functions for the hidden and output layers, respectively. Each layer multiplies its inputs by its weights, adds the layer's bias, and passes the result through its activation function to calculate the layer's output, as described in (1) – (2). The number of neurons in the hidden layer is represented by N, which can influence the network's capacity to learn complex patterns in the data. The number of neurons in the input layer matches the number of input features, while the number of neurons in the output layer depends on the network's application. In this study, one neuron is used at the output layer as the network forecasts a vector of PV output for the day ahead. Here, ω, b, f_H, f_y, N, input data split, learning rate, batch size and training function are learnable parameters calculated through the iterative training process, as outlined in Table 1. In training, the network's parameters are adjusted iteratively to minimize the gap between predicted and actual outputs, using optimization algorithms like stochastic gradient descent or its variants. Here, the Levenberg-Marquardt algorithm is employed (An Introduction to Neural Networks, 2022). To assess the training performance, an objective

Table 1
Parameters for FFNN.

Training function	Data split	N	f_h	f_y	Learning rate	Batch size
Levenberg-Marquardt	70/20/10	20	ReLU	Linear function	0.001	56

function is defined, which is the mean squared error (MSE) between the target value and the model output. MSE is a differentiable function, making it a good choice for training neural networks with gradient-based backpropagation for parameter optimization. Also, the squaring operation enhances its sensitivity to errors between predicted and actual values, potentially leading to more precise PV point forecasts post full network training.

Below is an outline of the steps taken to train the FFNN and adjust weights and biases for each layer.

a. The input data is fed into the FFNN. Each layer applies a set of weights and biases to the input data, followed by an activation function, to produce an output.
b. Calculate MSE between the predicted PV output and the PV actual output.
c. Calculate the gradient of the MSE with respect to weights and biases in the network.
d. Update weights and biases in the opposite direction of the gradient using the Levenberg-Marquardt algorithm.
e. Repeat steps a – d until the stopping criterion is met.

2.1.2. Prediction error analysis

While neural networks can generate PV point forecasts, they cannot be solely relied upon for scenario generation, as they are unable to capture sudden changes in PV output. Hence, here, the test data set is fed into the trained FFNN, and the gaps between the actual PV and predicted values are calculated for each time slot across different days. This is necessary because the range of prediction error varies across different hours for different days, Fig. 2. In other words, the prediction error for PV values at night is minimized because the network effectively learns that PV generation is zero during that time. However, daytime predictions are challenging due to input data volatility, making accurate PV output prediction impossible. Neglecting this aspect leads to inaccurate scenarios for PV output. Therefore, a time series of the mean and standard deviation of errors for every quarter-hour is calculated and fed into a normal distribution function to generate random values. These values are added to the FFNN's predicted PV output to create PV output scenarios. This process is repeated until a predetermined number of scenarios are generated. By considering the uncertainty specific to each time slot, this process effectively captures sudden changes in the PV output, improving the accuracy of scenarios. Consequently, the optimization model becomes more resilient in the face of such fluctuations, ensuring its ability to handle unexpected variations in PV output.

2.2. Scenario reduction

In scenario reduction, a subset of probabilistic PV power generation scenarios must be obtained without sacrificing the generality of the original set. In this regard, the literature contains two widely used iterative systemic methods; the forward selection algorithm and the backward reduction method (Gröwe-Kuska et al., 2003). In each iteration, the former selects a scenario with the highest probability that best represents the remaining scenarios and updates the scenario set. The selection process stops when the predetermined number of scenarios is chosen. The disadvantage of this method is that if only a few scenarios are chosen, some extreme cases with low probabilities may be ignored (Morales et al., 2009).

The selection process in the backward reduction method is based on the distance between each pair of scenarios, and one scenario is deleted from the scenario set in each iteration. The steps to deploy the backward reduction technique are as follows (Wu et al., 2007), assuming the total number of PV output scenarios is N_s and the objective number of scenarios is N_s^*.

a. Each PV uncertain power output scenario is equiprobable as $P(s_i) = 1/N_s$, and the initial number of the objective scenarios is $n = N_s$.

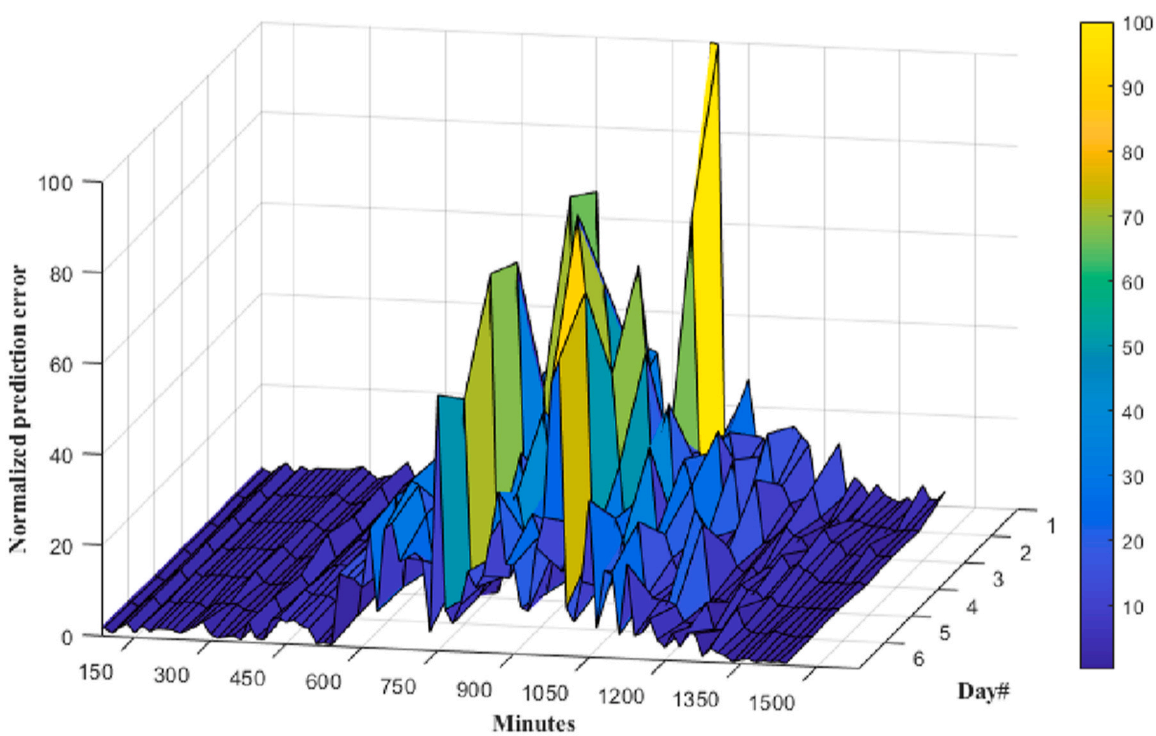

Fig. 2. Normalized range of prediction error for each time slot on a typical week.

b. Calculate the Kantorovich distance for each pair of scenarios (s_i, s_j) as follows.

$$D_k(s_i, s_j) = \left(\sum_{t=1}^{T} (P_{i,t}^{PV} - P_{j,t}^{PV})^2\right)^{1/2} \tag{3}$$

c. For each scenario s_i find the other nearest one s_r while $i \neq r$ and calculate the product of distance and scenario probability of each pair as follows.

$$PD_k(s_i, s_r) = D_k(s_i, s_r)P(s_i) \tag{4}$$

d. Repeat step c for every scenario, and compare the PD_k for all scenarios and find the pair with the minimum value. From this pair, delete the scenario with a smaller probability of occurrence s_d and update the probability of the remaining scenario s_r as;

$$P(s_r) = P(s_r) + P(s_d) \tag{5}$$

e. Update the number of remaining scenarios as $n = N_s - 1$ and repeat steps 2 – 4 until $n = N_s^*$.

Based on the Kantorovich metric, this method keeps a subset of scenarios that are closest to the original scenario set.

As explained above, this method systematically eliminates redundant scenarios, ensuring that essential scenarios are retained without overwhelming computational complexity. In simpler terms, using fewer scenarios in the optimization problem reduces computational burden and time while offering a comparable level of accuracy to solving the problem with the full set of scenarios.

3. Problem formulation

The scheduling problem presented in this work is formulated as a two-stage SO model. The first stage determines the optimal day-ahead power exchange with the main grid and BESS charge/discharge powers, which are fixed before the realization of random PV scenarios. These variables are known as the here-and-now variables. The second stage calculates recourse actions including the battery charge/discharge power, and power import/export from/to the main grid as wait-and-see variables. In the SO, here-and-now variables are determined before the realization of the scenario $s \in S$ and wait-and-see variables are calculated once the scenario s is known. More specifically, the second stage compensates for mismatches between the system state in scenario s and the day-ahead schedule that arise from PV uncertainties. It should be noted that only the PV is treated as an uncontrollable stochastic unit, and the BESS is a flexible available unit. Besides, tariffs and the household load profile are deterministic parameters, with the latter being a controllable variable for future research.

The objective function is formulated as a MILP model to minimize the expected day-ahead grid consumption and the battery degradation with a 15-minute time resolution. The target variables include grid import/export power, battery charge/discharge power, and the battery state of charge. In both day-ahead and real-time optimization, the battery can be charged from both the grid and local PV, and it can be discharged for on-site use and feeding the grid as well. Besides, real-time power exchange with the grid is allowed to correct any potential mismatches between the day-ahead schedule and the actual system state in real-time.

$$\begin{aligned} \min \sum_{t=1}^{T} & \left(P_{g,t}^{+,da}\pi_{TOU,t} - P_{g,t}^{-,da}\pi_{FIT}\right)\Delta t \\ & + \sum_{i=1}^{N_s^*} P(s_i) \sum_{t=1}^{T} (P_{g,i,t}^{+,rt}\pi_{TOU,t} - P_{g,i,t}^{-,rt}\pi_{FIT})\Delta t \end{aligned} \tag{6}$$

Subject to:

$$P_{g,t}^{+,da} + P_{g,i,t}^{+,rt} + P_{b,t}^{+,da} + P_{b,i,t}^{+,rt} + P_{pv,i,t} - P_{l,t} - P_{b,t}^{-,da} - P_{b,i,t}^{-,rt} - P_{g,t}^{-,da} - P_{g,i,t}^{-,rt} = 0 \tag{7}$$

$$P_{g,t}^{+,da} + P_{g,i,t}^{+,rt} \leq \alpha P_g^{+,\max} \tag{8}$$

$$P_{g,t}^{-,da} + P_{g,i,t}^{-,rt} \leq (1-\alpha) P_g^{-,\max} \tag{9}$$

$$P_{b,t}^{+,da} + P_{b,i,t}^{+,rt} \leq \beta P_b^{+,\max} \tag{10}$$

$$P_{b,t}^{-,da} + P_{b,i,t}^{-,rt} \leq (1-\beta) P_b^{-,\max} \tag{11}$$

$$P_{b,t}^{+,da} + P_{b,i,t}^{+,rt} + P_{pv,t} \leq P_{inv}^{\max} \tag{12}$$

$$SOC_t = SOC_{t-1} - \left(\frac{P_{b,t}^{+,da} + P_{b,i,t}^{+,rt}}{\eta_b^{+} E_b} - \frac{(P_{b,t}^{-,da} + P_{b,i,t}^{-,rt})\eta_b^{-}}{E_b}\right)\Delta t \tag{13}$$

$$(P_{b,t}^{-,da} + P_{b,i,t}^{-,rt} - P_{b,t}^{+,da} + P_{b,i,t}^{+,rt})\Delta t \leq (SOC^{\max} - SOC_{t-1})E_b \tag{14}$$

$$(P_{b,t}^{+,da} + P_{b,i,t}^{+,rt} - P_{b,t}^{-,da} + P_{b,i,t}^{-,rt})\Delta t \leq (SOC_{t-1} - SOC^{\min})E_b \tag{15}$$

$$\sum_{t=1}^{T} (SOC^{\max} - SOC_t) \leq \frac{TSOC^{\max}}{\varphi} \tag{16}$$

where t denotes a specific time within the total optimization horizon T; da and rt distinguish day-ahead parameters from real-time parameters, respectively; $P_g^{+/-}$ gives power flow imported/exported from/to the grid, considering $P_g^{+,\max}$, the load capacity of the home circuit breaker and $P_g^{-,\max}$, the contracted maximum power which can be sent to the grid. π_{TOU} and π_{FIT} are time-of-use (TOU) and feed-in-tariffs (FIT), respectively. The BESS charge/discharge power flow is denoted by $P_g^{-/+}$, and $P_b^{\max}$ denotes the maximum charge/discharge power of the BESS. The output of the PV is given by P_{pv} and P_l denotes the household load profile. In (8) – (11), α and β are binary variables that prevent simultaneous power import/export and BESS charge/discharge. The combined power from the PV and the BESS is limited by (12) to satisfy the hybrid PV/battery inverter's capacity, $P_{inv}^{\max}$.

The battery degradation control is expressed in (13) – (16). The dynamic of the battery state-of-charge (SOC) is calculated by (13), considering the battery capacity E_b, and its charge/discharge efficiency η_b, (Shirsat and Tang, 2021). The battery's SOC is kept within the allowable range using (14) – (15). The battery's cycles are controlled in (16), which is designed to reduce the number of deep damaging cycles (Rezaeimozafar et al., 2024). More specifically, the right side of (16) represents the total battery capacity that can be utilized for the dispatch strategy within the optimization horizon. This capacity can be modified using the tuning parameter, φ. The left side of (16) ensures that the battery operates in shallow cycles to prevent rapid depletion of the available capacity. In other words, by reducing the number of deep cycles, the left side allows for a more even distribution of the available capacity over the optimization period.

Constraints (7) – (16) are applied to both day-ahead and real-time optimization. The objective function is formulated as a MILP model, which can be studied in more detail in (Pochet and Wolsey, 2006). The flowchart of the proposed two-stage SO model is given in Fig. 3.

4. Results and Discussion

The proposed FFNN is trained using 15 months of real data (Dec 2018 – Feb 2020). This dataset is split into 70 % for training, 20 % for validation, and 10 % for testing. After training the network, the proposed two-stage optimization model is evaluated using real data collected on Apr 20th, 2020, from smart meters installed in an Irish case study (Lissa et al., 2021). The case study is a residential dwelling located

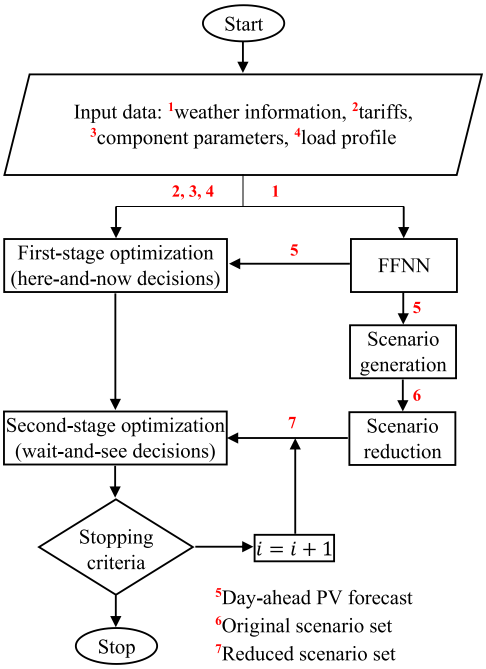

Fig. 3. Flowchart of the proposed BTM PV-BESS optimization model.

on the Island of Inis Mór, Co. Galway, Ireland. A schematic representation of the system under study is shown in Fig. 4. The dwelling was built in the 1970s and has a total floor area of 110 m^2. In recent years, the dwelling has undergone upgrades, including additional insulation to the walls and roof, as well as the installation of a PV panel array consisting of 8 panels with a nominal power of 2 kW. Additionally, a stationary battery energy storage system with a total capacity of 6 kWh was installed, bringing the total cost of the PV-BESS installation to approximately €8000 after the SEAI grant of €1600 is factored in (Solar Electricity PV Grants, 2024). Sensors in the dwelling measure indoor temperature, total electricity consumption, and total PV production, with the last two parameters being used in this work. Table 2 lists the specifications of the system under study. Electricity tariffs are based on a plan implemented by Electric Ireland, which our case study falls under. Simulations are run in MATLAB R2023a on a 7th generation Intel Core i7–1058 H 2.7Ghz Windows-based Dell Precision 3551 with 32 GB RAM specifications. The results are as follows.

Table 2
System Parameters.

PV system	
Technology	3.2 mm solar glass
Quantity of panels of the PV	8
Total production capacity (kW)	2
Possibility of curtailment	True
BESS	
Technology	Lithium Iron Phosphate
Total capacity (kWh)	6
Max charge power (kW)	5
Max discharge power (kW)	6
SOC range (%)	10 – 95
Inverter/Charger	
Technology	MultiPlus-II 48/3000/35–32
Max efficiency (%)	95
Max output power (kW)	5.5
Tariffs	
Grid consumption (¢/kWh)	29.62 (08:00 – 22:59), 15.24 (23:00 – 07:59), 8.23 (02:00 – 03:59)
Grid injection (¢/kWh)	5

4.1. FFNN

Accurate forecasts from the FFNN improve PV scenario generation, resulting in more efficient optimization solutions. Therefore, the performance of the proposed FFNN and its outputs are studied in Fig. 5 and Table 3 using mean squared error (MSE) and mean absolute error (MAE). MSE calculates the average squared deviation between predicted and actual values, while MAE calculates the average absolute deviation, offering clear insights into the model's performance. Lower MSE and MAE values indicate less deviation between predicted and true PV values, signaling more accurate forecasts.

Both the training and validation loss plots decrease to the point of stability, with the validation error slightly higher than the training error. Comparing the training error with the testing error, the same happens here. Both figures reach the stability point while there is a small gap between the training error and the testing error. To prevent the network from performing poorly on validation data while learning well on training data, training is stopped at epoch 50. To this end, if the validation performance degrades for 6 consecutive epochs, the training is stopped (early stopping); otherwise, continued training will most likely result in an overfit. It is important to note that the choice of 6 consecutive epochs is not a hard rule but rather a heuristic and may vary depending on the dataset, network architecture, and problem at hand. In

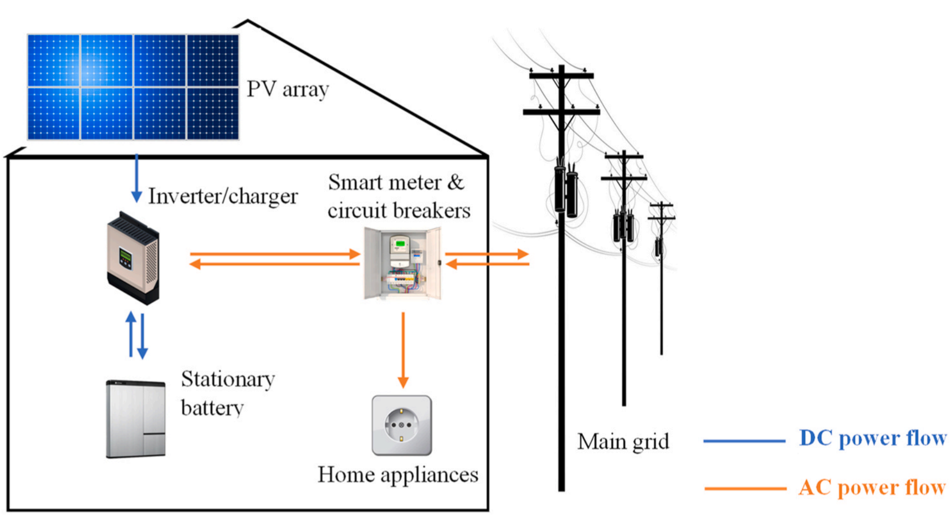

Fig. 4. Overview of the system under study.

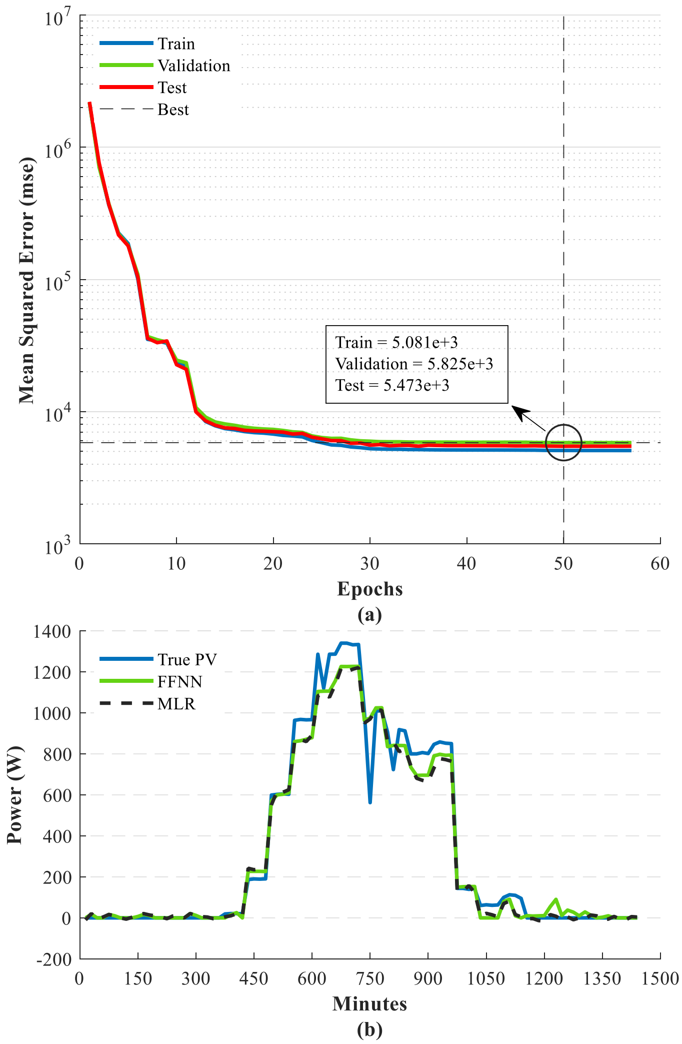

Fig. 5. The performance of the FFNN for PV output prediction, a) mean squared error analysis and b) true PV output vs FFNN vs multiple linear regression (MLR).

Table 3
FFNN performance on test data comparison vs MLR.

Model	Mean absolute error (MAE)	Mean squared error (MSE)
FFNN	17.85	5.473e+3
MLR	24.57	6.163e+3

this work, the threshold of 6 consecutive epochs is chosen based on monitoring the network's training performance to keep a balance between underfitting and overfitting.

Comparing the FFNN's predicted value to the real PV output in Fig. 5(b), the algorithm cannot predict sudden changes in the PV output, but it can estimate the PV's overall behavior, which is good enough for the scenario generation step. As only one hidden layer exists in the FFNN architecture, its accuracy is compared to that of a multiple linear regression model in Fig. 5(b) and Table 3.

The MLR model is trained for each time slot using the $M = 41$ features discussed in Section 2.1.1 plus historical PV outputs to capture the unique characteristics of each period during the day. The general MLR equation for predicting day-ahead PV output can be expressed as follows: where $C_{0,t}$ is the intercept for time t, $C_{1,t}$ to $C_{M+1,t}$ are the coefficients for the corresponding features and historical PV output for time slot t, and e_t is the error term for the prediction at time t. The regression model estimates the best values of $C_{0,t}$ to $C_{M+1,t}$ leading to the least error e_t.

$$P^{MLR}_{pv,t} = C_{0,t} + \sum\nolimits_{i=1}^{M+1} C_{i,t} x_{i,t} + e_t \tag{17}$$

Accordingly, the MLR can forecast day-ahead PV output with almost the same level of accuracy as the FFNN. However, the prediction here is based on input features with no uncertainties, which explains the accurate performance of the MLR. According to (AlShafeey and Csáki, 2021), unlike FFNN, the MLR outputs alter significantly when there are uncertainties in the input features, which is the case in real-world applications. Therefore, since the provided model will be upgraded and used in future studies on dynamic systems with real-time stochastic input features, the FFNN has been preferred over MLR for this work.

4.2. Scenario generation & reduction

The total number of generated scenarios and the reduced scenario set are shown in Fig. 6. First, 200 probabilistic PV scenarios are generated using the method described in Section 2.1. The scenario set is then reduced to 15 scenarios while preserving the key features of the original scenario set. It's worth noting that having more scenarios improves accuracy, but it also increases the computational burden for the SO.

Comparing the true PV output to the reduced scenario set, scenarios capture almost all variations in the PV output. Hence, the BESS scheduling model should be able to withstand sudden changes in PV output.

Given the importance of scenario set effects on optimization solutions in SO, and the need to maintain accuracy when reducing scenarios, we compare the chosen model from (Wu et al., 2007) employed in this work with three different techniques in Table 4. The comparison includes random sampling from the original scenario set (Jesmani et al., 2020), K-means clustering (Shi et al., 2021), and a fast forward selection model (Daneshvar et al., 2020). The random sampling model is selected to show the impact of random scenario selection on reduced set accuracy. The K-means-based model is chosen due to its widespread application in literature, while the forward reduction model is included to assess its efficacy compared to a backward reduction technique. To measure the models' accuracy, the mean values of the original set, μ_{org}, and the mean values of the reduced scenario set, μ_{red}, are employed in (18), calculating the difference relative to the means of the original set. Ideally, these means should be close, indicating that the reduction technique has preserved the central tendency accurately.

It should be noted that for a fair comparison, the random sampling is run five times and the mean values are calculated considering values

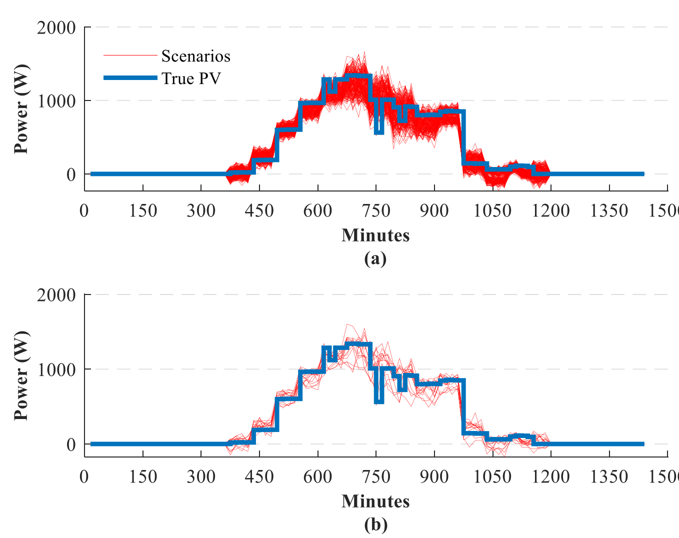

Fig. 6. The generated scenarios, a) The original scenario set versus the true PV output, and b) the reduced scenario set versus the true PV output.

Table 4
Results from comparing different models for scenario reduction.

	$\frac{\lvert\mu_{org}-\mu_{red}\rvert}{\lvert\mu_{org}\rvert}$			(18)
Model	(Wu et al., 2007)	(Jesmani et al., 2020)	(Shi et al., 2021)	(Daneshvar et al., 2020)
Accuracy (%)	79	37	44	73
Processing time (sec)	2	0.5	3	1.7

from each run. It is relatively easy to implement and fast, but it fails to maintain key features of the original set and results in inaccurate modeling. The k-means clustering model is easy to implement, but in this work, only PV time series data is available for clustering, which means the clustering is based solely on PV values, leading to inaccurate results. While clustering can be effective for data sets with more features, for this study, it does not provide reliable results. The forward reduction technique and the model used in this study are more complex but yield higher accuracy. However, as mentioned in Section 2.2, forward techniques may discard extreme scenarios with low probability, affecting the reduced set's mean values. Consequently, it shows slightly lower accuracy than the model employed in this study.

4.3. Impact of PV power output modelling

Table 5 shows the numerical results of the system simulation in four different case studies.

Case 1. The day-ahead PV output is accurately predicted, and a one-stage optimization problem is solved, assuming no uncertainty in the input data of the problem.

Case 2. The FFNN provides day-ahead point forecasts for PV output, which is used to solve a one-stage optimization problem with uncertainties introduced by the prediction errors of the FFNN.

Case 3. A day-ahead PV output profile is estimated using a Monte Carlo simulation method to solve a one-stage optimization problem with uncertainties caused by the prediction errors of the Monte Carlo model.

Case 4. The day-ahead BESS control problem is formulated as a two-stage SO problem incorporating probabilistic PV output scenarios to handle uncertainties, using the method proposed in this study.

The following formulations are used to obtain the PV output forecasts using a Monte Carlo simulation with 5000 iterations (Solar Electricity, 2022).

$$P(r_t) = \frac{1}{B(m_t, n_t)}\left(\frac{r_t}{r^{\max}}\right)^{m_t-1}\left(1-\frac{r_t}{r^{\max}}\right)^{n_t-1} \tag{19}$$

$$B(m_t, n_t) = \frac{\Gamma(m_t)\Gamma(n_t)}{\Gamma(m_t)+\Gamma(n_t)} \tag{20}$$

$$m_t = \mu_t\left(\frac{\mu_t(1-\mu_t)}{\sigma_t^2}-1\right) \tag{21}$$

$$n_t = (1-\mu_t)\left(\frac{\mu_t(1-\mu_t)}{\sigma_t^2}-1\right) \tag{22}$$

$$P_{pv,t} = \eta_{pv}P_{pv}^{\max}\bar{r}_t\left(1-0.004\left(T_{amb,t}-25\right)\right) \tag{23}$$

where $P(r_t)$ is the probability of a given solar irradiance r being realized at time t. B is the beta distribution function with m_t and n_t parameters. Γ denotes the gamma function. The mean and standard deviation of historical solar irradiance data, respectively, are μ and σ. Finally, (23) calculates the PV output power considering $\bar{r}$ obtained from iterative Monte Carlo simulations, PV system efficiency η_{pv}, maximum PV power capacity $P_{pv}^{\max}$, and ambient temperature T_{amb}.

According to Table 5, Case 4 produces the most similar results to Case 1. Case 3 has the lowest bill cost, but the solution is not practical because the PV output used in the calculations differs highly from the real values. In other words, the Monte Carlo simulation can almost estimate uncertain parameters with a normal distribution in an iterative process based on the law of large numbers (Graham and Talay, 2013). But, using a Monte Carlo method to predict PV behavior that changes abruptly does not provide accurate solutions to the BTM PV-BESS problem. Case 2 has the highest reported electricity bill, but when compared to Case 3, it provides more comparable outcomes to Case 1. Comparing Cases 2 and 4, scenario generation is required to enhance solutions and using only FFNN point forecasts will not produce the best results. Table 5 also highlights the problem of SO's long computation time by demonstrating that Case 4 takes 13 times longer to solve than the three other scenarios. This number could be 160 if the scenario reduction technique was not implemented, emphasizing the scenario reduction importance. However, it may appear unnecessary to discuss computational effort for a non-sensitive unit, i.e., a residential house. Nevertheless, the method is demonstrated for a domestic building based on the availability of data, but the same approach could be applied to installations on time-sensitive commercial/industrial buildings. It is worth noting that the results for Case 4 are calculated considering the outcomes from the first stage optimization and the weighted mean values of the results from scenarios in the second stage.

4.4. Charge/discharge schedules for BTM BESS

As for Case 4, Fig. 7 shows the optimal charge/discharge powers for the BTM battery derived from the first stage optimization. Recourse actions resulting from the second stage optimization are given in Fig. 8. According to Fig. 8, BESS charge/discharge and grid import/export recourse decisions compensate for mismatches between predicted PV power and generated scenarios. In other words, implementing the second-stage actions leads to more accurate BESS planning, as probabilistic PV scenarios are generated to capture the PV system's uncertain behavior. It also explains why the results from Case 4 in Table 5 are closer to Case 1 than the results from the other two case studies. For each scenario depicted in Fig. 8, recursive actions from the second stage optimization are given in Table 6.

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Table 5
Results were obtained from different case studies.

	Case 1	Case 2	Case 3	Case 4
Bill cost (€)	-0.5	0.4	-0.09	-0.2
Grid-to-load (Wh)	1523	1497	1348	1522
Grid-to-BESS (Wh)	0	0	0	48
PV-to-load (Wh)	2279	2409	2644	2335
PV-to-BESS (Wh)	1421	1421	1326	1447
PV-to-Grid (Wh)	4281	3865	3641	4206
BESS-to-load (Wh)	1710	1710	1624	1755
BESS-to-Grid (Wh)	0	0	0	0
Grid independence (%)	71.01	73.34	76.00	72.84
Processing time (sec)	5	5	5	65
Battery cycles	0.34	0.34	0.32	0.36

4.5. Comparative analysis

Table 7 compares the performance of the proposed two-stage SO model to that of four benchmark algorithms on data collected on April 20, 2020, from smart meters. The first is a rule-based model currently in use on-site, charging the battery to its maximum capacity during the

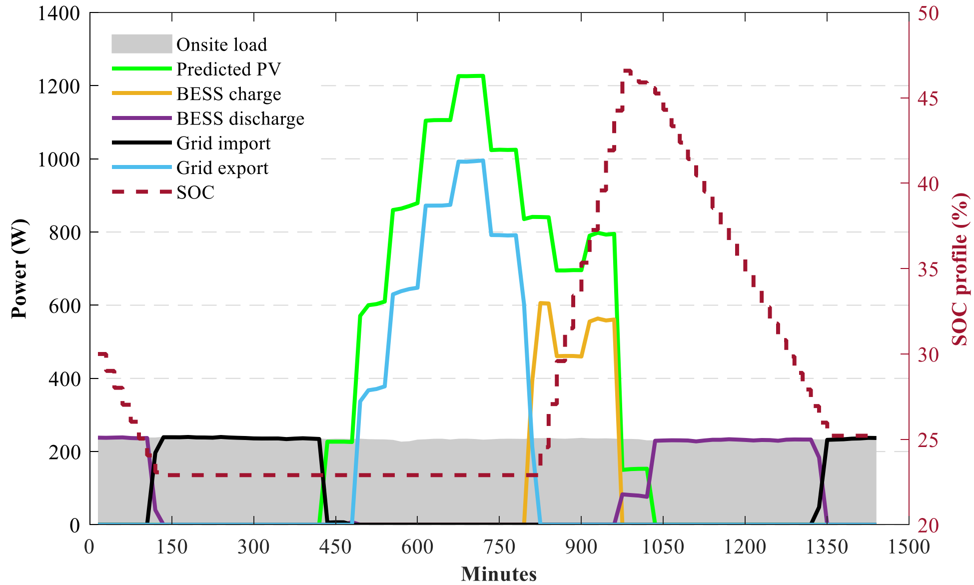

Fig. 7. Optimal solutions were obtained from the first-stage optimization.

Fig. 8. Recourse actions were obtained from the second-stage optimization.

Table 6
Numerical results from Fig. 8.

Scenario	Grid-to-load (Wh)	Grid-to-BESS (Wh)	PV-to-BESS (Wh)	BESS-to-load (Wh)	BESS-to-grid (Wh)	Min SOC (%)	Max SOC (%)
1	60	441	163	973	0	10.5	38.6
2	0	0	0	175	253	22.3	56.1
3	0	0	0	150	278	22.3	58.6
4	0	0	0	147	213	22.3	53.6
5	0	0	0	256	171	22.2	53.5
6	0	0	0	202	226	23.1	55.4
7	0	0	0	154	273	22.3	58.1
8	0	0	0	295	133	22.3	57.8
9	91	514	67	952	0	10.4	38.2
10	66	759	28	1144	0	10.2	40.6
11	0	0	0	117	311	22.3	56.8
12	0	0	0	133	295	22.3	58.3
13	0	0	0	119	308	22.3	59.3
14	0	0	0	172	255	24.5	58.4
15	59	453	37	869	0	10.1	38.8

night-boost period and discharging it during on-peak hours only when there is not enough PV. This aims to assess the proposed SO model in comparison to heuristic price-sensitive models. To compare our SO model with RO-based strategies, the second one, WRDDP from (Zhang et al., 2020), is a robust data-driven dynamic programming model. It considers PV output, electricity prices and the on-site load demand as stochastic parameters and selling energy back to the grid is not allowed. Meanwhile, to ensure a fair comparison, we only consider the PV output as stochastic, while assuming grid injection is possible. To assess how considering battery degradation impacts the problem solutions, the third one is a two-stage SO model proposed in (Kazemzadeh et al., 2019). The authors introduced a battery control technique based on a method of PV scenario generation, but they overlooked battery degradation in their problem formulation. The fourth work (Chen et al., 2012) introduces a method for generating stochastic scenarios, applied here for generating PV scenarios. Then, the optimization problem is solved using (6) – (16), with results given in Table 6, demonstrating the scenario's impacts on solutions. The results from Case 1 in Table 5 are also reported here to measure the effectiveness of the algorithms under study.

The Rule-based Model Is Faster and Easier to implement compared to our model. It monitors the system's state in real time before making decisions, which makes it reliable. However, its results differ significantly from the ideal case, Case 1, where our model outperforms the rule-based strategy. Also, it offers a high electricity bill as it fully charges the battery at night when it is not needed, which also puts unnecessary pressure on the battery, shortening its lifespan. The bill cost offered by WRDDP is identical to that of our model, but, the problem-solving time is over 18 times longer than ours. Also, there is a considerable difference between its outcomes and those of Case 1, particularly in terms of the BESS capacity discharged for on-site use. The WRDDP solves the problem following RO strategies, which rely on a worst-case scenario of estimation of PV output. This has an impact on decision-making accuracy, and may not be optimal for the proposed case study. The method of (Kazemzadeh et al., 2019) offers considerable results; however, in terms of solution accuracy and battery cycles, it exhibits poorer performance compared to the model proposed in this study. This is mainly because of the quality of the implemented PV scenarios and the oversight of battery degradation in the optimization problem, affecting BESS control. The optimization based on the method described in (Chen et al., 2012) for PV scenario generation produces the least favorable solutions. This can be attributed to the errors in the generated scenarios when compared to the actual PV values, highlighting the impacts of scenarios on the problem at hand. Overall, thanks to the accurate forecasts of the FFNN and the effective scenario generation method, our model outperforms its counterparts and offers a remarkably accurate solution, compared to the ideal case with error-free data available.

The electricity bill for the household under study without a PV-battery is calculated to be €2.53. In terms of long-term economic impacts, the payback period for the PV-BESS installation under the rule-based strategy stands at 14.1 years. Ideally, this duration can be halved to 7.2 years, with our model and the model outlined in (Zhang et al., 2020) offering a comparative payback period of 8 years. The longest payback period, reported for (Chen et al., 2012), exceeds that of the rule-based model, highlighting how inaccurate scenarios can adversely affect solutions. It is important to note that the payback period is simply calculated by dividing installation costs by the savings on bill costs. However, considering the improvement in battery cycles, our proposed model can also save long-term costs associated with battery degradation and replacement.

To gain deeper insights into the importance of optimal design for FFNN, we compare our design's performance for day-ahead PV forecasting to that of FFNNs proposed in (Kaffash et al., 2021; Vagropoulos et al., 2016). This comparison is presented in Table 8, focusing on the mean absolute error value and considering the impacts of the size of available data. Notably, as the specific design details of the networks in (Kaffash et al., 2021; Vagropoulos et al., 2016) are not provided, this comparison is based on separate parameter tuning conducted individually for each network in the present study.

The proposed FFNN achieves remarkable performance with a limited dataset, thanks to incorporating both environmental feature set and historical PV output values. In other words, the model's predictions are informed not just by past PV data but also by environmental factors.

Table 8
MSE for various periods compared between FFNN, (Kaffash et al., 2021) and (Vagropoulos et al., 2016).

PV predictive model	6 months	12 months	18 months	24 months
FFNN	64.6	42.7	17.5	18.1
(Kaffash et al., 2021)	127.8	89.3	65.4	51.3
(Vagropoulos et al., 2016)	284.2	211.7	271.9	267.1

Table 7
Results from comparing the presented model with three benchmarks.

	Case 1	Two-stage SO	Rule-based	(Zhang et al., 2020)	(Kazemzadeh et al., 2019)	(Chen et al., 2012)
Bill cost (€)	-0.5	-0.2	0.98	-0.2	-0.1	1.7
Grid-to-load (Wh)	1523	1489	1683	1145	1367	2576
Grid-to-BESS (Wh)	0	48	3889	0	106	12
PV-to-load (Wh)	2279	2302	2382	2366	2349	1958
PV-to-BESS (Wh)	1421	1447	0	1760	1612	1164
PV-to-Grid (Wh)	4281	4206	5702	3698	3900	3648
BESS-to-load (Wh)	1710	1722	1447	2001	1796	974
BESS-to-Grid (Wh)	0	0	0	0	0	0
Grid independence (%)	72.3	73	69.4	79.2	75.2	53.2
Processing time (sec)	5	65	2	580	59	112
Battery cycles	0.34	0.36	0.57	0.42	0.41	0.24
Payback period (year)	7.2	8	14.1	8	8.3	26.4

However, this expanded input scope also exposes the model to noise within the dataset, a challenge addressed during data preparation. Nevertheless, providing the model with 24 months of data slightly diminishes its accuracy, primarily due to heightened volatility within the data sets, especially in weather-related input data. Nonetheless, its accuracy remains significantly superior to the two other designs. Another notable finding is that the accuracy of (Kaffash et al., 2021) increases with the availability of more data. This improvement is attributed to the fact that the predictions solely rely on historical PV output. Hence, when the network is exposed to a larger number of samples, it can achieve higher accuracy. However, even with access to two years of data, it fails to achieve the same level of accuracy as the model discussed in this study.

5. Conclusion

This paper presents a two-stage scenario-based SO model for the optimal BTM PV-BESS control problem. A FFNN predicts day-ahead PV output, based on which 200 probabilistic PV scenarios are generated using effective prediction error analysis to achieve highly accurate PV scenarios. A backward scenario reduction technique is then used to preserve only the 15 most probable ones, cutting down on computing time. The first stage optimization solves the day-ahead BESS scheduling problem formulated as a MILP. The second stage calculates recourse actions to compensate for mismatches between the day-ahead scheme and the system state in real-time. The problem's objective function is designed to keep the expected daily bill costs and battery degradation at minimum rates.

As shown, feeding the two-stage SO model with highly accurate PV scenarios from the proposed scenario generation model provides solutions nearly identical to those from the ideal scenario with perfect PV estimation available. In other words, accurate scenario generation along with second-stage recourse decisions can compensate for errors in day-ahead PV estimates, maximizing the accuracy of solutions. Besides, the given approach saves 76 % more on bill costs than the on-site rule-based model, cutting the system payback period by over 6 years. The WRDDP model and our model both calculate the same electricity bill costs, however, our model's suggested solutions regarding battery charge/discharge closely align with the ideal case and require 88.8 % less computation time. Also, the results confirm that employing the FFNN along with the error analyzing technique allows for achieving high accuracy even with relatively small data sets. This aspect holds significant importance as it makes our model applicable to case studies where access to long-term historical PV data for training accurate predictive models is limited or unavailable. Moreover, the method's performance remains consistent regardless of system size and its geographical location, adaptable based on scenario set size for accuracy and computational efficiency. The proposed model can be adapted for systems under various policies and regulations by modifying the problem formulation in (6) – (16) to comply with regulatory requirements.

While the proposed model is reliable, its accuracy is highly dependent on the quality and quantity of initial scenarios. More scenarios yield more accurate PV modelling, resulting in more accurate solutions. But, this increases the size of the problem and thus computational burden and resource needs. Also, this method requires continuous generation of scenarios and computation of solutions, particularly when frequent optimization runs with short intervals are required. This raises concerns regarding operational costs and may limit the implementation of this method for residential units, where cost minimization takes precedence. For example, the approx. €3000 cost of the computing system used in this study could extend the payback period by almost 3 years, affecting the households' interest in the proposed model. Introducing a cloud computing system to handle calculations for end-users could serve as a solution, albeit it necessitates support from policymakers and utilities and may currently pose financial challenges. Future research will focus on model-free algorithms, bypassing dependency on scenarios and uncertainty modeling for fast, cost-effective decision-making. Training reinforcement learning agents to optimize PV-BESS control stands out as a promising approach.

Funding

This work was supported by the European Regional Development Fund (ERDF) under the Interreg North West Europe (NWE) project STEPS.

Author Statement

We the authors declare that this manuscript is original, has not been published before and is not currently being considered for publication elsewhere. We confirm that the manuscript has been read and approved by all named authors and that there are no other persons who satisfied the criteria for authorship but are not listed. We further confirm that the order of authors listed in the manuscript has been approved by all of us. We understand that the Corresponding Author is the sole contact for the Editorial process. He is responsible for communicating with the other authors about progress, submissions of revisions and final approval of proofs.

CRediT authorship contribution statement

Maeve Duffy: Writing – review & editing, Supervision, Funding acquisition. **Rory F.D. Monaghan:** Writing – review & editing, Supervision. **Mostafa Rezaeimozafar:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Data curation, Conceptualization. **Enda Barrett:** Writing – review & editing, Validation, Supervision, Formal analysis.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Mostafa Rezaeimozafar reports financial support was provided by European Regional Development Fund.

Data availability

Data will be made available on request.

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